

# Molecular Photocopying

## How Polymerase Chain Reaction (PCR) Amplifies the Code of Life

### THE BIG IDEA

**PCR selectively amplifies a specific DNA sequence, turning a small genetic signal into enough material for analysis.**

Think of it as a molecular photocopier: identify one sentence in an enormous library, then repeatedly copy it until it becomes easy to study.

DNA  
TARGET

1

### AMPLIFY

Specific target DNA

2

### ANALYSE

Gel, qPCR or sequencing

3

### APPLY

Diagnostics • AMR • forensics

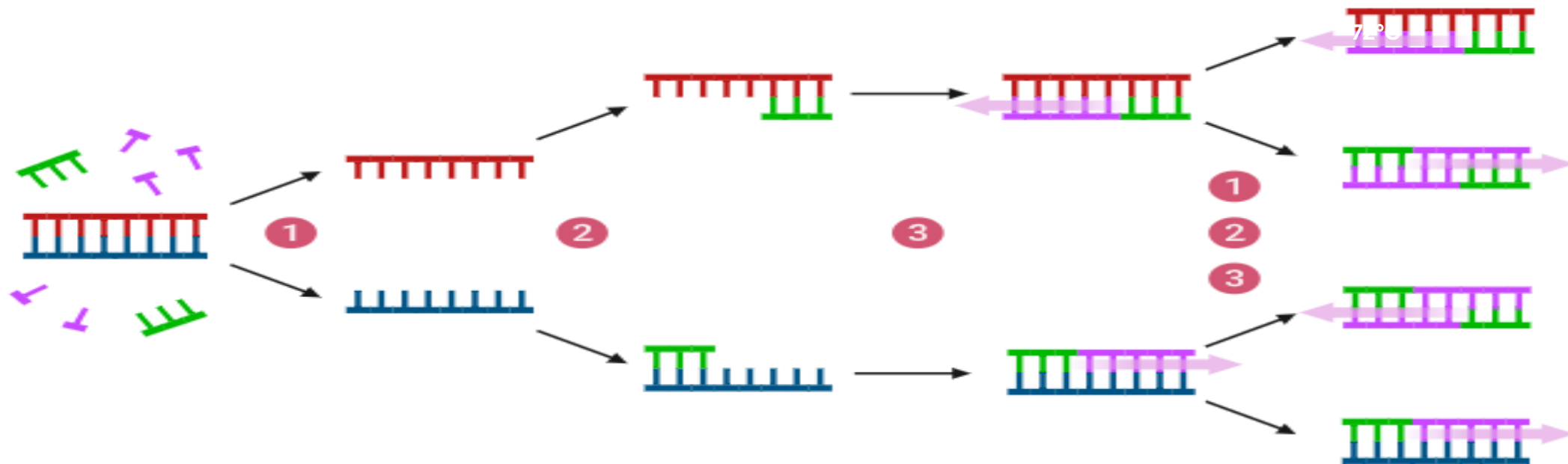
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# The PCR cycle: three controlled temperature steps

A thermal cycler repeatedly changes temperature so DNA can separate, primers can bind and polymerase can synthesise new strands.

## Polymerase chain reaction - PCR

Each cycle provides new templates for the next round.



### 1) Denaturing (95-96°C)

Heat disrupts the hydrogen bonds between complementary bases, separating double-stranded DNA into single templates.

### 2) Annealing (68°C)

Cooling allows short DNA primers to bind to complementary sequences flanking the target region. Primer sequence and temperature help determine specificity.

### 3) Elongation (72°C)

A thermostable DNA polymerase extends from the primers using dNTPs. Conventional PCR commonly uses Taq polymerase.

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# Exponential amplification and why specificity matters

In an ideal reaction, target DNA approximately doubles each cycle:  $2^n$ , where  $n$  is the number of cycles.



## WHAT MAKES PCR SPECIFIC?

|                            |                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------|
| <b>Primer pair</b>         | defines the target region and provides the starting point for synthesis.                  |
| <b>Reaction conditions</b> | including annealing temperature, help minimise non-specific amplification.                |
| <b>Controls</b>            | positive and negative controls help distinguish genuine amplification from contamination. |

## AFTER AMPLIFICATION

PCR products can be visualised by agarose gel electrophoresis, quantified in qPCR workflows or prepared for sequencing.

Real reactions are not perfectly efficient and eventually reach a plateau as reagents become limiting.

# Why PCR matters in biomedical science

01

## INFECTIOUS-DISEASE DIAGNOSTICS

Targeted amplification can identify pathogen nucleic acid sequences. RNA targets, such as SARS-CoV-2, require an additional reverse-transcription step before amplification.

02

## ANTIMICROBIAL-RESISTANCE RESEARCH

PCR can screen clinical or environmental samples for genes associated with antimicrobial resistance, including targets investigated in water and wastewater surveillance.

03

## FORENSIC DNA PROFILING

Amplification enables analysis of selected DNA regions from very small biological samples, supporting genetic profiling.

04

## GENETIC & GENOMIC RESEARCH

PCR supports cloning, sequencing preparation and targeted genetic analysis, often forming one stage of a larger experimental workflow.

# From technique to reliable biomedical evidence

## WHAT THIS TECHNIQUE DEMONSTRATES

-  **Sequence specificity** Primers define what is amplified.
-  **Experimental control** Positive and negative controls support interpretation.
-  **Reproducibility** Temperature, reagent choice and cycle number affect performance.
-  **Downstream analysis** Amplified products must be visualised, quantified or sequenced.

## KEY TAKEAWAY

**PCR is not simply a way to make “more DNA”. It a sequential amplification system linking molecular biology, experimental design and downstream analysis.**

Understanding primer specificity, thermal cycling, polymerase activity and controls is fundamental to using PCR reliably in a biomedical laboratory.

## KEY REFERENCES

- Mullis, K. et al. (1986). Specific enzymatic amplification of DNA in vitro: the polymerase chain reaction. Cold Spring Harbor Symposia on Quantitative Biology, 51, 263–273.
- Saiki, R.K. et al. (1988). Primer-directed enzymatic amplification of DNA with a thermostable DNA polymerase. Science, 239(4839), 487–491.

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